

Chapter 14: Introduction to Inference

The Practice of Statistics in the Life Sciences
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Part III: Foundations of Inference

From Sampling Distributions to Inference

Chapter 13 explained how sample statistics vary from one random sample to another.

Chapter 14 uses those sampling distributions to answer a new question:

What can a sample tell us about an unknown population parameter?

Statistical inference connects sample evidence to population conclusions while accounting for random sampling variation.

Two Main Goals of Inference

Two major forms of inference are introduced in this chapter.

- A **confidence interval** estimates an unknown population parameter and describes the precision of the estimate.
- A **hypothesis test** evaluates how compatible the observed data are with a specific claim about a population parameter.

Both methods depend on the sampling distributions developed in Chapter 13.

Simple Conditions for Introductory Inference

The first procedures in this chapter use a simplified setting so that the reasoning is easy to see.

1. The observations come from an appropriate random sampling process and can be treated as independent.
2. The sampling distribution of $\bar{x}$ can be modelled by a Normal distribution.
3. The population standard deviation σ is treated as known.

Later chapters replace the known value of σ with the sample standard deviation s and use Student's t distribution.

Why Start with a Known Population Standard Deviation?

Knowing the true value of σ is unusual in real applications.

The assumption is useful here because it separates two ideas:

- The **logic of inference** can be learned first.
- The additional mathematics required when σ must be estimated can be introduced later.

The study design still matters. A correct formula cannot repair serious bias caused by an inappropriate sample.

A sample statistic can be used as an estimate of a population parameter.

For a population mean μ , the natural point estimate is

$$\bar{x}$$

A point estimate gives one number. A confidence interval adds a margin of error so that sampling uncertainty is also described.

Examples 14.1 and 14.2: Mean Height

Question

A health survey provides a sample of $n = 217$ eight-year-old boys from the population of interest. The sample mean height is $\bar{x} = 132.5$ cm. For this introductory example, assume the population standard deviation is $\sigma = 10$ cm.

How can the sample mean be used to estimate the unknown population mean height μ while accounting for random sampling variation?

Examples 14.1 and 14.2: Sampling Distribution

Explanation

The sampling distribution of $\bar{x}$ has centre μ and standard deviation

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

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For this sample size,

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This quantity is the **standard error** of the sample mean under the model.

Examples 14.1 and 14.2: Approximate 95% Distance

Explanation

The 68–95–99.7 rule says that approximately 95% of a Normal distribution lies within two standard deviations of its mean.

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$$2(0.679) \approx 1.36 \text{ cm}$$

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Using the exact standard error gives

$$2(0.679) \approx 1.36 \text{ cm}$$

The textbook-style calculation rounds the standard error to 0.7 cm first, which gives an approximate margin of error of

$$2(0.7) = 1.4 \text{ cm}$$

Examples 14.1 and 14.2: Confidence Interval

Solution

The lower endpoint is

$$132.5 - 1.4 = 131.1$$

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$$132.5 - 1.4 = 131.1$$

The upper endpoint is

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The approximate 95% confidence interval is

$$(131.1, 133.9) \text{ cm}$$

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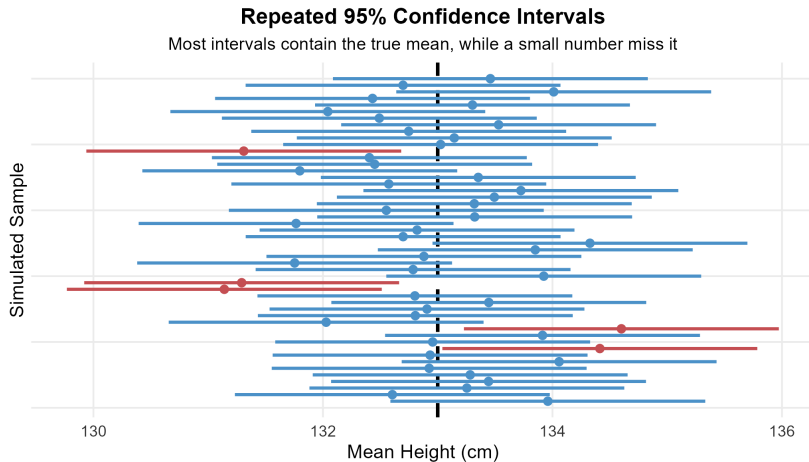
$$132.5 + 1.4 = 133.9$$

The approximate 95% confidence interval is

$$(131.1, 133.9) \text{ cm}$$

We are approximately 95% confident that the population mean height lies between 131.1 cm and 133.9 cm because this method captures the true mean in about 95% of repeated samples under the stated model.

Repeated 95% Confidence Intervals



What Does 95% Coverage Mean?

The population mean is fixed. The intervals change from sample to sample.

A 95% confidence procedure has a **long-run coverage rate of 95%** under the model.

This means that approximately 95% of intervals produced by the method would contain the true parameter over many repeated random samples.

A small simulation does not need to show exactly 95% coverage. The theoretical statement concerns the long-run procedure.

Anatomy of a Confidence Interval

Most confidence intervals have the form

$$\text{estimate} \pm \text{margin of error}$$

Three ideas determine the interval:

- The **estimate** determines the centre.
- The **standard error** describes sampling variability.
- The **confidence level** determines how wide the interval must be for the desired long-run coverage.

The Estimate

When estimating a population mean, the centre of the interval is usually the sample mean.

$$\bar{x}$$

For the height example,

$$\bar{x} = 132.5 \text{ cm}$$

The sample mean is unbiased for μ under the usual random-sampling model. This property does not guarantee that one particular sample mean is close to the population mean.

The Margin of Error

For a confidence interval based on the Normal distribution,

$$m = z^* \frac{\sigma}{\sqrt{n}}$$

The margin of error describes uncertainty caused by random sampling under the model. It does not automatically account for other sources of error such as poor measurement, nonresponse or selection bias.

The Confidence Level

The confidence level C describes the long-run success rate of the interval-building procedure.

- A 90% interval uses a smaller critical value than a 95% interval.
- A 99% interval uses a larger critical value than a 95% interval.

Higher confidence requires a wider interval when the sample size and population standard deviation remain unchanged.

Precision and Confidence

Two changes have predictable effects on interval width.

Larger Sample Size

A larger sample reduces the standard error because

$$\frac{\sigma}{\sqrt{n}}$$

becomes smaller as n increases.

Higher Confidence Level

A higher confidence level uses a larger z^* value, which makes the interval wider.

How to Interpret a Confidence Interval

A suitable interpretation is:

95% Confidence

We are 95% confident that the population parameter lies between the reported endpoints because the procedure used to construct the interval captures the true parameter in 95% of repeated samples under the stated conditions.

The confidence level describes the reliability of the **method** over repeated sampling.

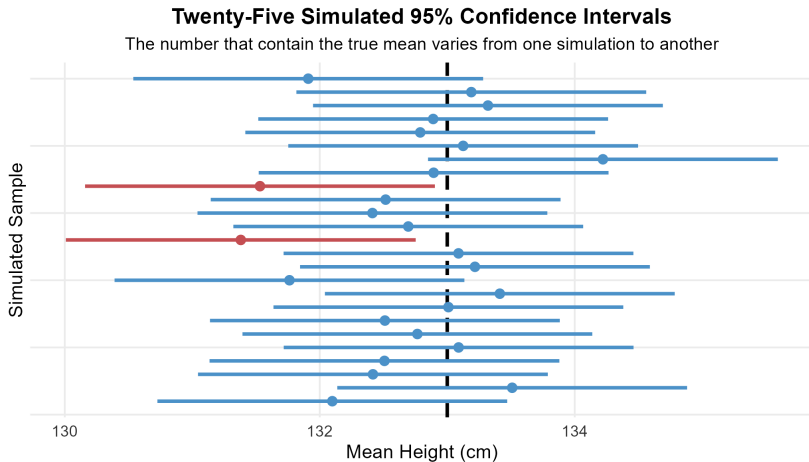
What 95% Confidence Does Not Mean

After an interval has been calculated, the population parameter is fixed.

The interval either contains the parameter or it does not. The difficulty is that the parameter is unknown.

For a frequentist confidence interval, we do not say that there is a 95% probability that the fixed parameter lies inside the specific interval that was already calculated.

Twenty-Five Simulated 95% Confidence Intervals



Common Critical Values

The critical value z^* is chosen so that the central area under the Standard Normal curve equals the desired confidence level.

Confidence Level	Critical Value z^*
90%	1.645
95%	1.960
99%	2.576

Confidence Interval for a Population Mean

Under the introductory conditions with known σ ,

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

The same formula can be viewed as

$$\text{estimate} \pm \text{critical value} \times \text{standard error}$$

The estimate determines the centre. The standard error measures sampling variation. The critical value sets the desired confidence level.

The Four-Step Process for Confidence Intervals

A consistent process keeps the calculation connected to the scientific question.

1. **STATE:** The population question and parameter are identified.
2. **PLAN:** The confidence level and appropriate procedure are selected.
3. **SOLVE:** The conditions are assessed before the interval is calculated.
4. **CONCLUDE:** The interval is interpreted in the context of the population.

Example 14.3: Leaf Surface Area

Question

A botanist wants to estimate the mean surface area of mature *Monstera deliciosa* leaves in a greenhouse population. A random sample contains 31 mature leaves. The population standard deviation is treated as known at $\sigma = 15 \text{ cm}^2$.

What is a 95% confidence interval for the population mean leaf surface area?

Example 14.3: Observed Leaf Data

114	100	104	89	102	91	114	114
103	105	108	130	120	132	111	128
118	119	86	72	111	103	74	112
107	103	98	96	112	112	93	

Example 14.3: Observed Leaf Data

114	100	104	89	102	91	114	114
103	105	108	130	120	132	111	128
118	119	86	72	111	103	74	112
107	103	98	96	112	112	93	

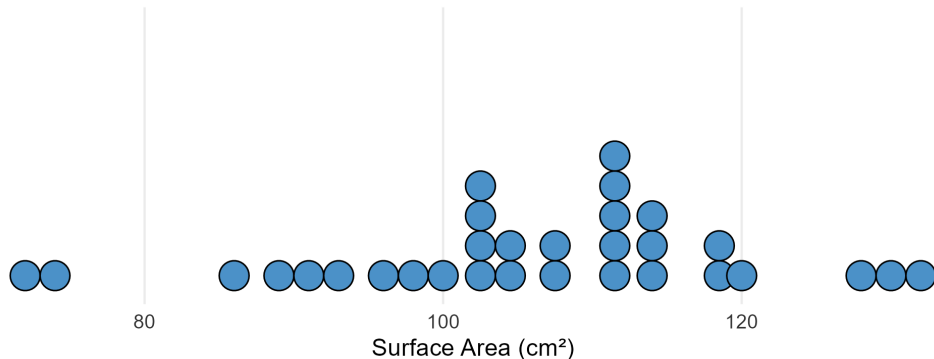
Explanation

The parameter is the population mean leaf surface area μ . The first step is to assess whether the sampling model is reasonable before calculating the interval.

Example 14.3: Leaf Surface Area Distribution

Dotplot of Leaf Surface Area

The sample is slightly left-skewed with no extreme outliers



Example 14.3: Checking the Conditions

Explanation

The sample was described as randomly selected, which supports the sampling condition.

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With $n = 31$, the Central Limit Theorem provides additional support for an approximately Normal sampling distribution of $\bar{x}$ when the population has finite variance.

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The population standard deviation is treated as known at

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Example 14.3: Calculating the Interval

The sample mean is

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$$\bar{x} \approx 105.84$$

The standard error is

$$SE = \frac{15}{\sqrt{31}} \approx 2.69$$

The 95% critical value is

$$z^* = 1.960$$

The margin of error is

$$m = 1.960(2.69) \approx 5.28$$

Example 14.3: Confidence Interval Solution

Solution

The interval is

$$105.84 \pm 5.28$$

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Therefore,

$$(100.56, 111.12)$$

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The interval is

$$105.84 \pm 5.28$$

Therefore,

$$(100.56, 111.12)$$

We are 95% confident that the population mean surface area of mature leaves in the greenhouse population is between approximately 100.6 cm^2 and 111.1 cm^2 .

Example 14.3: Confidence Interval in R

```
# The 31 observed leaf surface areas are stored in a vector.
leaf_data <- c(
  114, 100, 104, 89, 102, 91, 114, 114,
  103, 105, 108, 130, 120, 132, 111, 128,
  118, 119, 86, 72, 111, 103, 74, 112,
  107, 103, 98, 96, 112, 112, 93
)

# The population standard deviation is treated as known.
sigma_leaf <- 15
xbar_leaf <- mean(leaf_data)
n_leaf <- length(leaf_data)

# qnorm(0.975) gives the 95% Normal critical value.
z_star_leaf <- qnorm(0.975)
se_leaf <- sigma_leaf / sqrt(n_leaf)
margin_leaf <- z_star_leaf * se_leaf

c(xbar_leaf - margin_leaf, xbar_leaf + margin_leaf)
```

Looking Beyond the Simple Conditions

The known- σ assumption is mainly a teaching device.

In most real studies, the population standard deviation is unknown.

Later chapters replace σ with the sample standard deviation s and use Student's t distribution.

The central logic remains the same: an estimate is combined with a standard error and an appropriate critical value.

From Estimation to Hypothesis Testing

A confidence interval begins with an unknown parameter and asks for a plausible range of values.

A hypothesis test begins with a specific population claim and asks whether the observed sample data are compatible with that claim.

The central question is:

If the null hypothesis were true, how unusual would the observed sample result be?

Examples 14.4 and 14.5: Phosphorus Levels

Question

In the general adult population, blood phosphorus levels are modelled as Normal with mean 1.2 mmol/L and standard deviation 0.1 mmol/L. Researchers suspect that the mean may be lower among adults aged 75 to 79.

A retrospective chart review provides measurements from 12 patients. Does the sample provide evidence that the population mean for this older group is below 1.2 mmol/L?

Examples 14.4 and 14.5: Phosphorus Data

1.26	1.39	1.00	1.00	1.00	1.10
0.87	1.23	1.19	1.29	1.03	1.18

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$$\bar{x} \approx 1.1283 \text{ mmol/L}$$

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0.87	1.23	1.19	1.29	1.03	1.18

The sample mean is

$$\bar{x} \approx 1.1283 \text{ mmol/L}$$

The sample mean is below 1.2, but sampling variation alone can produce differences between $\bar{x}$ and μ .

Examples 14.4 and 14.5: Hypotheses

Explanation

The test compares the observed difference with the variation expected if the general-adult mean still applied to the older population.

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Explanation

The test compares the observed difference with the variation expected if the general-adult mean still applied to the older population.

Solution

The null hypothesis is

$$H_0 : \mu = 1.2$$

The alternative hypothesis is

$$H_a : \mu < 1.2$$

The alternative is one-sided because the scientific question concerns a decrease.

The Null and Alternative Hypotheses

The **null hypothesis** gives the specific population value or model being evaluated.

$$H_0 : \mu = \mu_0$$

The **alternative hypothesis** gives the competing population claim that is supported when the data provide sufficient evidence against H_0 .

Hypotheses refer to population parameters. They do not describe sample statistics that have already been observed.

One-Sided and Two-Sided Alternatives

A lower-tail alternative has the form

$$H_a : \mu < \mu_0$$

An upper-tail alternative has the form

$$H_a : \mu > \mu_0$$

A two-sided alternative has the form

$$H_a : \mu \neq \mu_0$$

The direction should be determined by the scientific question before the sample result is used to choose the test.

Example 14.6: Aspirin Tablet Manufacturing

Question

A pharmaceutical process is designed to produce aspirin tablets with a population mean active-ingredient amount of 325 mg. Both underfilling and overfilling are undesirable. What hypotheses should be used to evaluate whether the population mean differs from the target?

Example 14.6: Choosing the Hypotheses

Explanation

The reference value is 325 mg. The scientific concern includes departures above or below the target.

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Solution

The null hypothesis is

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Explanation

The reference value is 325 mg. The scientific concern includes departures above or below the target.

Solution

The null hypothesis is

$$H_0 : \mu = 325$$

The alternative hypothesis is

$$H_a : \mu \neq 325$$

This is a two-sided test because deviations in either direction matter.

The Logic of a P-Value

A hypothesis test begins by temporarily assuming that H_0 is true.

The null sampling distribution then tells us which sample results are common and which are unusual.

A result that is very unusual under the null model provides evidence against that model.

The Test Statistic

For a one-sample z test of a population mean,

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

The numerator measures the observed difference from the null value.

The denominator is the standard error expected from random sampling.

The resulting z value tells us how many standard errors the observed sample mean is from the null mean.

The P-Value

Definition

The P-value is the probability, calculated under the assumption that the null hypothesis and model conditions are true, of obtaining a test statistic at least as extreme as the observed statistic in the direction or directions specified by the alternative hypothesis.

A P-value is not the probability that H_0 is true.

A P-value is not the probability that the observed result happened only because of chance.

Interpreting P-Values

- A small P-value provides stronger evidence against H_0 because the observed result is unusual under the null model.
- A large P-value means the observed result remains reasonably compatible with H_0 .
- Failure to reject H_0 does not prove that H_0 is true.

Statistical significance does not measure the practical or biological importance of an effect.

Example 14.7: One-Sided Phosphorus Test

Question

For the phosphorus study,

$$H_0 : \mu = 1.2$$

$$H_a : \mu < 1.2$$

with $n = 12$, $\bar{x} \approx 1.1283$ mmol/L and known $\sigma = 0.1$ mmol/L.

What is the one-sided P-value?

Example 14.7: Standard Error

Explanation

Under H_0 , the sample mean has centre 1.2 mmol/L.

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The standard error is

$$SE = \frac{0.1}{\sqrt{12}} \approx 0.0289$$

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Explanation

Under H_0 , the sample mean has centre 1.2 mmol/L.

The standard error is

$$SE = \frac{0.1}{\sqrt{12}} \approx 0.0289$$

The observed sample mean is lower than the null mean, so the next step is to measure that difference in standard-error units.

Example 14.7: Test Statistic and Tail

The test statistic is

$$z = \frac{1.1283 - 1.2}{0.0289} \approx -2.48$$

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Example 14.7: Test Statistic and Tail

The test statistic is

$$z = \frac{1.1283 - 1.2}{0.0289} \approx -2.48$$

The alternative hypothesis is

$$H_a : \mu < 1.2$$

Therefore, the P-value is the Standard Normal area to the **left** of the observed z statistic.

Example 14.7: P-Value Solution

Solution

$$P(Z \leq -2.48) \approx 0.0065$$

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Under the null model, a sample mean of approximately 1.1283 mmol/L or lower would occur in about 0.65% of repeated samples of size 12.

Example 14.7: P-Value Solution

Solution

$$P(Z \leq -2.48) \approx 0.0065$$

Under the null model, a sample mean of approximately 1.1283 mmol/L or lower would occur in about 0.65% of repeated samples of size 12.

The data provide strong evidence against H_0 under the stated assumptions and support the conclusion that the population mean is below 1.2 mmol/L.

Example 14.7: One-Sided Test in R

```
# The observed phosphorus measurements are stored in mmol/L.
phosphorus <- c(
  1.26, 1.39, 1.00, 1.00, 1.00, 1.10,
  0.87, 1.23, 1.19, 1.29, 1.03, 1.18
)

# The null mean and population standard deviation define the model.
mu_null_phos <- 1.2
sigma_phos <- 0.1

xbar_phos <- mean(phosphorus)
n_phos <- length(phosphorus)
se_phos <- sigma_phos / sqrt(n_phos)
z_phos <- (xbar_phos - mu_null_phos) / se_phos

# The alternative is lower-tailed, so pnorm() gives the left-tail area.
p_phos <- pnorm(z_phos)
p_phos
```

Example 14.8: Two-Sided Aspirin Test

Question

A manufacturing process targets $\mu = 325$ mg with known $\sigma = 5$ mg. A random sample of $n = 10$ tablets has mean $\bar{x} = 326.9$ mg.

The hypotheses are

$$H_0 : \mu = 325$$

$$H_a : \mu \neq 325$$

What is the two-sided P-value?

Example 14.8: Standard Error and Test Statistic

Explanation

The standard error is

$$SE = \frac{5}{\sqrt{10}} \approx 1.581$$

Example 14.8: Standard Error and Test Statistic

Explanation

The standard error is

$$SE = \frac{5}{\sqrt{10}} \approx 1.581$$

The test statistic is

$$z = \frac{326.9 - 325}{1.581} \approx 1.202$$

Example 14.8: Standard Error and Test Statistic

Explanation

The standard error is

$$SE = \frac{5}{\sqrt{10}} \approx 1.581$$

The test statistic is

$$z = \frac{326.9 - 325}{1.581} \approx 1.202$$

The two-sided alternative requires equally extreme outcomes in both tails.

Example 14.8: Two-Sided P-Value Solution

Solution

For the symmetric Standard Normal distribution,

$$2P(Z \geq 1.202) \approx 0.2295$$

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A P-value of about 0.2295 is not small.

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Solution

For the symmetric Standard Normal distribution,

$$2P(Z \geq 1.202) \approx 0.2295$$

A P-value of about 0.2295 is not small.

The data do not provide statistically significant evidence that the population mean differs from 325 mg. This conclusion does not prove that the process mean is exactly 325 mg.

Example 14.8: Two-Sided Test in R

```
# The null value is the target tablet amount.
mu_null_aspirin <- 325
sigma_aspirin <- 5
n_aspirin <- 10
xbar_aspirin <- 326.9

# The z statistic measures the standardized distance from the target.
se_aspirin <- sigma_aspirin / sqrt(n_aspirin)
z_aspirin <- (xbar_aspirin - mu_null_aspirin) / se_aspirin

# abs() lets the same two-sided formula work for positive or negative z.
p_aspirin <- 2 * pnorm(-abs(z_aspirin))
p_aspirin
```

Significance Levels α

The significance level is a decision threshold chosen before the analysis.

The usual decision rule is

- When $P \leq \alpha$, the result is statistically significant and H_0 is rejected.
- When $P > \alpha$, the result is not statistically significant and H_0 is not rejected.

Common values include

$$\alpha = 0.05$$

$$\alpha = 0.01$$

What Does α Mean?

For a correctly calibrated test, α controls the long-run probability of a Type I error when the null hypothesis is true.

The statement

$$\alpha = 0.05$$

does not mean that there is a 5% probability that H_0 is true.

The word **significant** also does not mean that an effect is large or biologically important.

The One-Sample z Test for a Mean

Under the introductory conditions, the null hypothesis is

$$H_0 : \mu = \mu_0$$

The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

The test statistic measures the observed difference from the null value in units of the standard error.

Which Tail Gives the P-Value?

The alternative hypothesis determines the relevant tail or tails.

- For $H_a : \mu > \mu_0$, the P-value is the area to the right of the observed z value.
- For $H_a : \mu < \mu_0$, the P-value is the area to the left of the observed z value.
- For $H_a : \mu \neq \mu_0$, the P-value includes equally extreme outcomes in both tails.

For a two-sided z test in R,

$$P = 2 \text{pnorm}(-|z|)$$

Example 14.9: Body Temperature

Question

An historical reference value for mean human body temperature is 98.6°F . A later study reports oral temperatures from $n = 130$ healthy adult volunteers with sample mean $\bar{x} = 98.25^{\circ}\text{F}$. For this introductory exercise, assume the observations can be treated as independent and representative of the population being studied. Also assume known $\sigma = 0.6^{\circ}\text{F}$. Do the data provide evidence that the population mean differs from 98.6°F ?

Example 14.9: Plan and Standard Error

Explanation

The question concerns a difference in either direction, so a two-sided test is appropriate.

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$$H_0 : \mu = 98.6$$

$$H_a : \mu \neq 98.6$$

Example 14.9: Plan and Standard Error

Explanation

The question concerns a difference in either direction, so a two-sided test is appropriate.

$$H_0 : \mu = 98.6$$

$$H_a : \mu \neq 98.6$$

The standard error is

$$SE = \frac{0.6}{\sqrt{130}} \approx 0.0526$$

Example 14.9: Test Statistic

The test statistic is

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The observed sample mean is about 6.65 standard errors below the null mean.

A result at least this far from 98.6 in either direction would be extremely unusual under the null model.

Example 14.9: P-Value and Conclusion

Solution

The two-sided P-value is approximately

$$2.9 \times 10^{-11}$$

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Solution

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The P-value is extremely small. Under the stated assumptions, the data provide extremely strong evidence against H_0 .

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Solution

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$$2.9 \times 10^{-11}$$

The P-value is extremely small. Under the stated assumptions, the data provide extremely strong evidence against H_0 .

The result is not impossible under H_0 . Its probability is simply extremely small. The strength of the population conclusion also depends on whether the study sample appropriately represents the population of interest.

Example 14.9: Body Temperature Test in R

```
# The null model uses the historical reference mean.
mu_null_temp <- 98.6
xbar_temp <- 98.25
n_temp <- 130
sigma_temp <- 0.6

# The standard error and z statistic describe the observed difference.
se_temp <- sigma_temp / sqrt(n_temp)
z_temp <- (xbar_temp - mu_null_temp) / se_temp

# The alternative is two-sided, so both tails are included.
p_temp <- 2 * pnorm(-abs(z_temp))

c(Standard_Error = se_temp, Z_Statistic = z_temp, P_Value = p_temp)
```

The Confidence Interval and Test Connection

Two-sided confidence intervals and two-sided hypothesis tests are closely connected when they use the same model.

A test at significance level α corresponds to a confidence level

$$C = 1 - \alpha$$

For example,

$$\alpha = 0.05$$

corresponds to

$$C = 0.95$$

Using a Confidence Interval to Test a Null Value

Under the same two-sided model,

- A null value outside the corresponding confidence interval leads to rejection of H_0 .
- A null value inside the corresponding confidence interval leads to failure to reject H_0 .

This equivalence requires the same standard error, significance level and two-sided model.

Example 14.10: Body Temperature Confidence Interval

Question

Using $\bar{x} = 98.25^\circ\text{F}$, $n = 130$ and known $\sigma = 0.6^\circ\text{F}$, what is the 95% confidence interval for the population mean?

How does the interval lead to the same conclusion as the two-sided test of

$$H_0 : \mu = 98.6$$

at $\alpha = 0.05$?

Example 14.10: Calculating the Interval

Explanation

The 95% critical value is

$$z^* = 1.960$$

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The standard error is

$$SE \approx 0.0526$$

Example 14.10: Calculating the Interval

Explanation

The 95% critical value is

$$z^* = 1.960$$

The standard error is

$$SE \approx 0.0526$$

The margin of error is

$$m = 1.960(0.0526) \approx 0.1031$$

Example 14.10: Confidence Interval Solution

Solution

The interval is

$$98.25 \pm 0.1031$$

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The interval is

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Therefore,

$$(98.15, 98.35)$$

Example 14.10: Confidence Interval Solution

Solution

The interval is

$$98.25 \pm 0.1031$$

Therefore,

$$(98.15, 98.35)$$

The null value 98.6°F lies outside the 95% confidence interval. The corresponding two-sided test rejects H_0 at the 5% significance level.

Example 14.10: Confidence Interval in R

```
# qnorm(0.975) gives the critical value for a 95% interval.
z_star_temp <- qnorm(0.975)

# The confidence interval uses the same standard error as the test.
margin_temp <- z_star_temp * se_temp
lower_temp <- xbar_temp - margin_temp
upper_temp <- xbar_temp + margin_temp

round(
  c(Lower_Bound = lower_temp, Upper_Bound = upper_temp),
  2
)
```

Chapter 14 Summary

- Statistical inference uses sample data to learn about population parameters while accounting for sampling variation.
- Confidence intervals combine an estimate with a margin of error.
- A P-value measures how unusual the observed evidence would be under the null model.
- The significance level α provides a decision threshold chosen before the analysis.
- Two-sided confidence intervals and two-sided tests are directly connected when they use the same statistical model.

A Consistent Inference Workflow

A first-year inference problem becomes easier when the same reasoning is used each time.

1. The population parameter and scientific question are identified.
2. The study design and assumptions are assessed.
3. The sampling distribution and standard error are identified.
4. The confidence interval or test is calculated.
5. The numerical result is interpreted in context.
6. The conclusion respects both statistical evidence and the limitations of the study design.